INDEPENDENT STUDY — LIVING SUPERVISOR REPORT

physio-placebo

Is the LLM reading the signal, or is the verbalizer doing the classifying?

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This edition	14 October 2026
Closed through	Week 8 (git tags `week-01-done` … `week-06-done`)
Next block	Week 9 (verbalizer ablation) — not started
Repository	`https://github.com/shrijacked/physio-placebo`
Pre-registration	`https://osf.io/62r5t` DOI 10.17605/OSF.IO/62R5T

This is the document I will keep sending. Each time a week is tagged done, I add that week's real numbers and leave later weeks blank. Nothing here changes the filed OSF text.

1 Revision log

Edition	Closed through	What this PDF added
21 Sep 2026	Week 6	First living edition. Weeks 1–6 numbers, A/B/C tables, methods sketch, SDS, input examples.
11 Oct 2026	Week 6	Health-LLM comparison arm. Prompting and fine-tuning are separate tables. Weeks 7–8 are still not started.
14 Oct 2026	Week 8	Surrogate ladder and classical ledger. 120 model cells, raw logprobs saved. Week 9 not started.

2 Snapshot

Block	State	One-line result
Week 1 — data	Closed 16 Sep	WESAD 15/15, MAUS 22/22, CogWear 10/11; 0 feature-NaN drops
Week 2 — floors	Closed 16 Sep	WESAD strong (logreg 0.918); MAUS and CogWear weak
Week 3 — anchor + OSF	Closed 20 Sep	MedAlpaca MAE 2.689 ± 0.038 vs paper 0.76 ± 0.1
Weeks 4–5 — A/B	Closed 21 Sep	Best LLM cell: Qwen WESAD A bin-mean 0.595 [0.550, 0.647]
Week 6 — plots (C)	Closed 21 Sep	Qwen2.5-VL at chance on WESAD and MAUS
Comparison arm	Added 11 Oct	PMData stress MAE only. Separate from the WESAD grid.
Weeks 7–8 — surrogates	Closed 14 Oct	120 cells scored. WESAD logreg stays high on IAAFT (0.880) and on the label-preserving permutation (0.954).
Week 9 — verbalizer	Not started	Neutralize / mis-sign adjectives (the paper's core)
Week 10 — SDS card	Not started	Protected mid-semester result

What I would like you to take from this edition. Real-signal prompting is in. Classical WESAD is still far above every LLM cell. Paradigm C (plots) is at chance, which is the direction we pre-registered. The next scientific step is the surrogate ladder, not more prompt tuning.

3 The study, in one paragraph

Published wearable-LLM papers report that language models can infer stress or workload from sensor text. They almost never test whether that accuracy depends on the physiological signal. This study runs the same class of prompts on WESAD, MAUS, and CogWear under strict leave-one-subject-out, then replaces the recording with a ladder of surrogates and measures how much accuracy survives. The intellectual core is a verbalizer ablation: do models read the numbers, or the class-suggestive adjectives in the template? A negative result is pre-registered as publishable.

Scoring is locked: one forward pass, argmax over the log-probabilities of label tokens `A`/`B`. No free generation. Prompt IDs are chosen on two held-out subjects per dataset (seed 1337) and then frozen before any surrogate run.

The pre-registered Surrogate Dependence Score is

$$\mathrm{SDS} = \frac{M_{\mathrm{real}} - M_{\mathrm{surrogate}}}{M_{\mathrm{real}} - M_{\mathrm{chance}}},$$

with M the pooled LOSO macro-F1 and M_{chance} the permutation-floor mean. If a cell's real-signal margin over that floor has a bootstrap CI that includes zero, SDS is undefined and only the raw delta is reported.

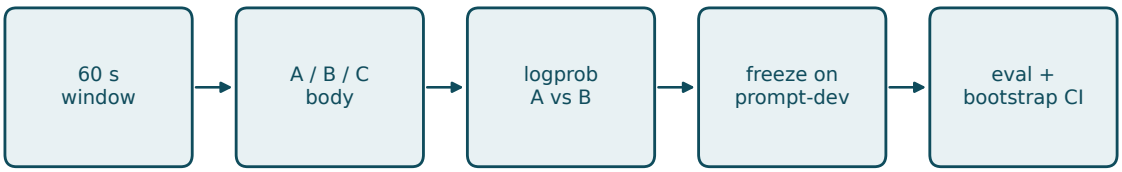


Figure 1: One cell. A 60 s window becomes an A, B, or C body; the scorer takes arg max of $P(\texttt{A})$ and $P(\texttt{B})$. Prompt IDs freeze on two prompt-dev subjects. Reported numbers are eval subjects only.

The same WESAD 60 s ECG window in the three locked forms. A is bin-mean (n=128), the winning WESAD A cell. B is verbalizer (i); four of eighteen feature sentences (the adjective leak is intentional and is the Week 9 ablation). C is the matplotlib PNG the VL model receives. Subject id and class are not on the plot.

A — bin-mean ECG (n=128)

```
ecg fs=700 Hz n=128 scheme=bin_mean:
0.04024 0.005173 -0.01352 -0.01912 0.03245 -0.00422 -0.02569 0.0196
... 117 values omitted ...
-0.04705 -0.00251 -0.06615
```

B — verbalizer (i), 4 of 18 feature sentences

```
eda_tonic_mean is reduced (0.7358), calmer-typical.
hrv_mean_hr is elevated (89.28), stress-typical.
hrv_sdnn is reduced (41.18), stress-typical.
hrv_rmssd is reduced (17.7), stress-typical.
```

4 Week 1 — data and harness

Windows are 60 s with a 30 s stride, cut strictly inside one protocol segment.

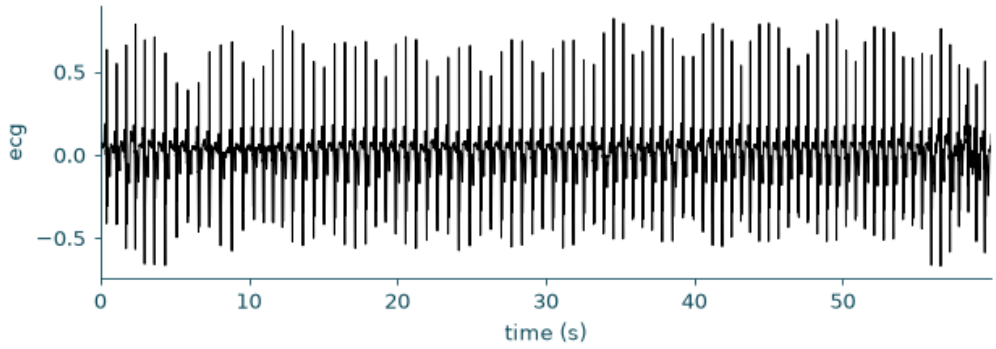


Figure 2: Paradigm C view of that same window. No subject id and no class name.

Dataset	Subjects	Windows	Class split	NaN drops
WESAD	15 / 15	1,042	731 non-stress / 311 stress	0
MAUS	22 / 22	1,086	362 low / 724 high	0
CogWear	10 / 11	114	39 baseline / 75 Stroop	0

CogWear subject 3 is excluded because PhysioNet is missing that subject's EDA file (logged in the prereg). Feature code is hash-locked (NeuroKit2 0.2.13). LOSO and a prompt-dev split are tested. SensorLM captioning templates are vendored (Apache-2.0) for a later optional verbalizer variant; they are not a model-under-test.

5 Week 2 — classical floor

Locked *before* any LLM grid run, on byte-identical feature matrices (seed 1337; 1,000 within-subject label permutations). These are the numbers an LLM cell must beat.

Dataset	Logreg L2	Depth-3 tree	Majority	Perm. p95	Signal
WESAD (n=15)	0.918	0.848	0.412	0.433	Strong
MAUS (n=22)	0.444	0.482	0.400	0.413	Weak
CogWear (n=10)	0.569	0.560	0.397	0.541	Weak

WESAD is the dataset that can actually test signal dependence. MAUS and CogWear barely clear (or fail) the permutation floor; many of those cells will hit the pre-registered SDS validity gate. We will then report raw deltas, not a ratio on a near-zero denominator.

6 Week 3 — fidelity anchor and OSF

OSF registration is public (`https://osf.io/62r5t`). Predictions on file:

- SDS < 0.3 for raw-series and plot paradigms;
- verbalized features lose $> 50\%$ of their above-chance margin when adjectives are neutralized;
- adversarially mis-signed adjectives flip $> 40\%$ of items.

Fidelity anchor (Health-LLM PMData stress, zero-shot MedAlpaca-7b) on the HTI RTX A6000 (20 Sep 2026): MAE $\mathbf{2.689 \pm 0.038}$ versus paper 0.76 ± 0.1 (gap +1.929; 6/897 parse failures). Completions are mostly "out of 5 / out of 10"; the locked first-number scorer reads 5 or 10. I am *not* retuning that scorer — the gap is the result. The 299-item eval split and the three prompt hashes were locked before the run.

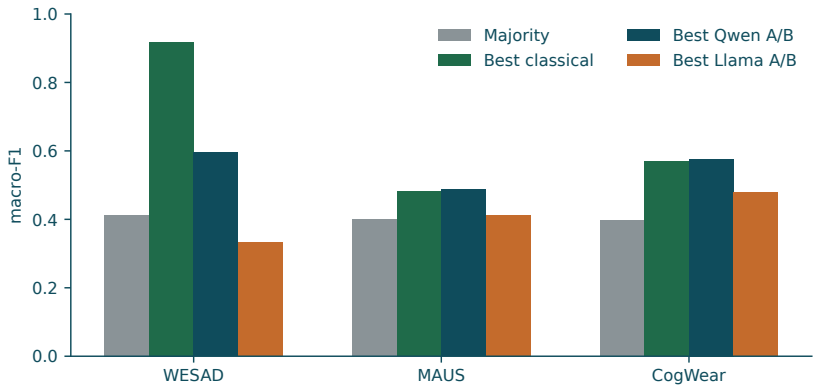


Figure 3: Locked classical floor versus the best real-signal A/B cell per model. WESAD is the only dataset where a classical model is far above majority. Every LLM bar is the best of that model's three A/B cells.

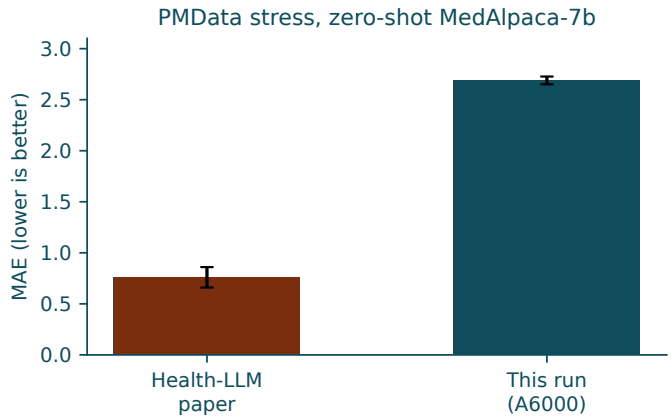


Figure 4: Health-LLM Table 3 cell (paper 0.76 ± 0.1) versus the locked first-number reproduction on the HTI A6000 (2.689 ± 0.038).

Executed models (public OSF update, 20 Sep 2026) are Qwen3-8B, Llama-3.1-8B-Instruct, and Qwen2.5-VL-7B-Instruct via vLLM in fp16. The filed Models section still names gpt-4.1; that is a disclosed post-registration deviation (OpenAI 50 req/day cap). I am not rewriting the frozen OSF Models paragraph.

7 Weeks 4–5 — Paradigms A and B

Prompt-dev subjects (excluded from eval): WESAD S2/S8, MAUS 015/023, CogWear 4/8. Scoring failures: 0 / 11,916 eval windows. CIs are subject-bootstrap, 1,000 resamples, seed 1337.

Dataset	Cell	Qwen3-8B	Llama-3.1-8B
WESAD	A stride	0.427 [0.392, 0.465]	0.333 [0.301, 0.370]
WESAD	A bin-mean	**0.595 [0.550, 0.647]**	0.230 [0.227, 0.233]
WESAD	B verbalizer	0.584 [0.462, 0.699]	0.283 [0.247, 0.324]
MAUS	A stride	0.467 [0.436, 0.499]	0.413 [0.403, 0.426]
MAUS	A bin-mean	0.483 [0.452, 0.516]	0.400 [0.400, 0.400]
MAUS	B verbalizer	0.487 [0.439, 0.531]	0.400 [0.400, 0.400]
CogWear	A stride	0.576 [0.443, 0.689]	0.404 [0.392, 0.417]
CogWear	A bin-mean	0.396 [0.380, 0.411]	0.440 [0.399, 0.495]
CogWear	B verbalizer	0.475 [0.393, 0.504]	0.479 [0.375, 0.640]

Winning prompts differ by cell (frozen in `configs/frozen_prompts.lock.yaml`). The bold cell is Qwen, WESAD, Paradigm A, bin-mean, clinical 0-shot — the only LLM number that clearly clears WESAD majority (0.412). It is still far below the classical logreg floor (0.918). Llama's two MAUS cells at 0.400 ± 0 are constant-class on every eval subject, not a crash. CogWear CIs are wide (n=90 eval windows).

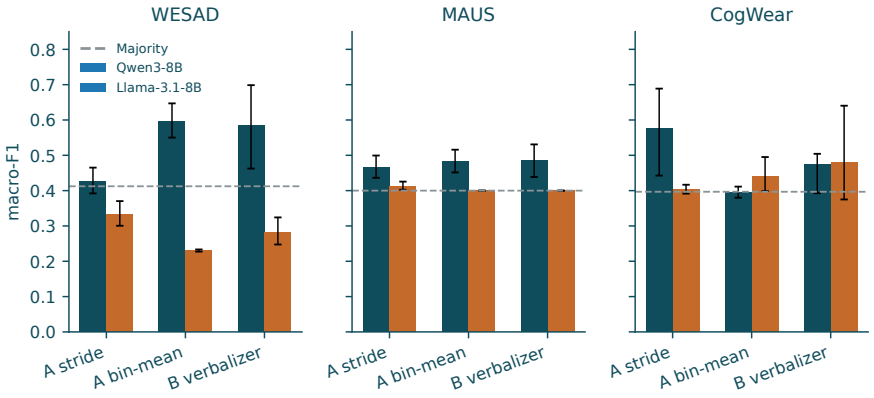


Figure 5: Real-signal Paradigms A and B, eval subjects only, subject-bootstrap 95% CIs. Dashed line is majority-class macro-F1. Qwen is above Llama on WESAD and MAUS. The WESAD A bin-mean cell is the only LLM number that clearly clears WESAD majority.

8 Week 6 — Paradigm C (plots)

Qwen2.5-VL-7B-Instruct, matplotlib line plot of the primary channel (WESAD ECG, MAUS PPG). Same window → same PNG bytes. No subject id and no class name on the figure. CogWear has no C cell. Frozen prompt: instruction 4-shot.

Dataset	Prompt	macro-F1 [95% CI]	Windows
WESAD	instruction 4-shot	0.412 [0.410, 0.414]	906
MAUS	instruction 4-shot	0.400 [0.399, 0.400]	990

Both sit on chance. The tight CIs mean the model is almost always emitting one class. That is consistent with pre-registered prediction 1 (plot paradigm should not carry signal). I am not retuning C prompts.

9 What is next (not started)

1. **Week 9.** Verbalizer ablation on Paradigm B: original / neutralized / mis-signed adjectives. SensorLM templates remain a pending yes/no from you (variant iv).

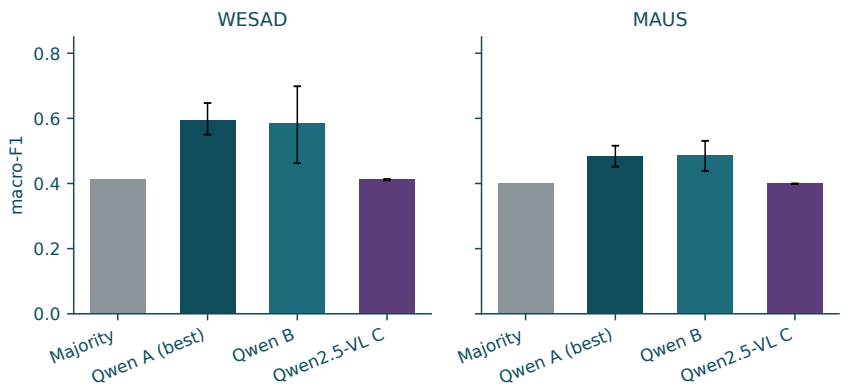


Figure 6: Paradigm C against majority and the same model's A/B cells. Qwen2.5-VL sits on chance on both WESAD and MAUS.

2. **Week 10.** SDS report card. Protected. If the calendar slips, we cut stretch work, not this.

10 Comparison arm — Health-LLM, PMData stress

This is not a new week. WESAD, MAUS, and CogWear macro-F1 stay in the tables above. The match is Kim et al., Health-LLM, CHIL 2024, PMData stress, MAE. Train rows are the released pool minus one sensor string that also appears in the locked 299-item eval (653 left). The paper's 10% test split is not claimed to be that eval. Recipe lock: `configs/healthllm_finetune.lock.yaml`.

Row	Model	Adaptation	MAE
Paper Table 3	MedAlpaca-7b	zero-shot	0.76 ± 0.1
Locked rerun	MedAlpaca-7b	zero-shot	2.689 ± 0.038
Different backbone	Qwen3-8B	zero-shot, same prompts	0.795 ± 0.020
Different backbone	Llama-3.1-8B-Instruct	zero-shot, same prompts	2654 ± 43

Qwen answers `Answer: 3`. Llama writes Python that pastes step counts; the locked first-number scorer reads those counts. The scorer was not changed. Always answering 3 on this eval has MAE 0.401, which is still below Qwen.

Row	Model	Adaptation	MAE
Paper Table 4	MedAlpaca-7b	fine-tuned on PMData	0.38
This arm, eval seed 0	MedAlpaca-7b LoRA	3 epochs, batch 128, fp16	2.697

297 of 299 fine-tune completions parsed. They are still "out of 5", so the scorer reads 5. The same model's zero-shot seed 0 on this eval was 2.714. The LoRA run did not move that cell. Deviations from the paper (no oversampling, warmup 0 because this pool is 653 rows, per-device batch 1) are in the lock file.

11 Weeks 7–8 — surrogate ladder

Every frozen prompt was scored again after the window was replaced. 120 cells, 0 scoring failures. Each row in `results/surrogates/eval/` keeps the label-token logprobs. The label-preserving permutation is reported and is not a null.

Classical logistic regression on the surrogate features (`results/surrogates/ledger/20261013T193152Z/ledge`

Surrogate	WESAD	MAUS	CogWear
Gaussian	0.412	0.400	0.486
IAAFT	0.880	0.405	0.458
Wrong subject, label broken	0.411	0.405	0.380
Flatline	0.412	—	0.397
Signal deleted	—	—	—
Wrong subject, label kept	0.954	0.494	0.773

A dash means the features were absent or entirely missing, so the regression was not fit. On WESAD the phase-randomized series still carries class information (0.880), and keeping the condition while swapping the person carries still more (0.954). Gaussian noise, a broken label, and a flat line sit on the majority floor (0.412). SDS is Week 10. It is not computed here.

12 Open items for you

Still useful from you. SensorLM verbalizer variant (iv): yes or no. Code is vendored; extra compute is near zero. A decision lets me freeze the Week 9 ablation set.

GPU access is resolved (HTI A6000, account `shrijak`). The surrogate grid already ran there.

Shrijak Kumar • 14 October 2026 • Working log: `PROGRESS.md` • This report does not change the filed OSF text.